

Titration report



Method

Method ID	SERCALKSAMPLES	Release state	Released
Name	TA of Seawater	Modified by	SystemInternal
Modified on	2/22/2024 12:07:06 PM		

Summary

Samples

Sample ID	Name	Result	State
15	FTS2023 8/8 1530 rep4, 15	Sample size	63.249 g ⁽¹²⁾

Main Result Details

Result title	Sample ID1
Result name	Sample ID2
Result	Result comment
	Result state
	Result state comment
	Nominal value
	Compliance state
	Is result deleted

Calculation formula

Raw Results

Scope 1/1, Sample 15/17 (FTS2023 8/8 1530 rep4; 15)

Sample (S1)

Creation time	2/22/2024 5:42:28 PM ⁽¹²⁾
Sample start time	2/22/2024 5:42:28 PM ⁽¹²⁾
Sample size	63.249 g ⁽¹²⁾
Correction factor	1.0 ⁽¹²⁾
Temperature	25.0 °C ⁽¹²⁾
Density	1.0 g/mL ⁽¹²⁾

Titration stand (Titrationstand1)

Creation time	2/22/2024 5:42:29 PM ⁽¹²⁾
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Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 5:42:29 PM ⁽¹²⁾
T	25.3 °C ⁽¹²⁾
E	7.796 pH ⁽¹²⁾
tMe	01:12 min ⁽¹²⁾
ESTMe	4.800 pH ⁽¹²⁾

Titration (EQP) [1] (TitrationEQP1)

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LabX

Creation time 2/22/2024 5:43:42 PM ⁽¹²⁾
EEQ1 4.575 pH ^(9,12)
EHN1 6.232 pH ⁽¹²⁾
Q1 0.123115 mmol ^(9,12)
VEQ1 1.229921 mL ^(9,12)
nEQ 2 ^(9,12)

Measure (normal) [1] Scope 1/1, Sample 15/17

Measured Values Scope 1/1, Sample 15/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.800	23.5	
2	5.069	23.7	0.269
3	5.313	23.9	0.244
4	5.510	24.2	0.197
5	5.678	24.4	0.168
6	5.827	24.6	0.149
7	5.962	24.8	0.135
8	6.075	24.9	0.113
9	6.191	25.0	0.116
10	6.298	25.0	0.107
11	6.400	25.1	0.102
12	6.494	25.1	0.094
13	6.582	25.2	0.088
14	6.666	25.2	0.084
15	6.747	25.2	0.081
16	6.816	25.2	0.069
17	6.887	25.2	0.071
18	6.949	25.3	0.062
19	7.009	25.3	0.060
20	7.068	25.3	0.059
21	7.123	25.3	0.055
22	7.168	25.3	0.045
23	7.213	25.3	0.045
24	7.256	25.3	0.043
25	7.296	25.3	0.040
26	7.333	25.3	0.037
27	7.369	25.3	0.036
28	7.402	25.3	0.033
29	7.429	25.3	0.027
30	7.458	25.3	0.029
31	7.484	25.3	0.026
32	7.509	25.3	0.025
33	7.533	25.3	0.024
34	7.554	25.3	0.021
35	7.574	25.3	0.020
36	7.591	25.3	0.017
37	7.609	25.3	0.018
38	7.626	25.3	0.017

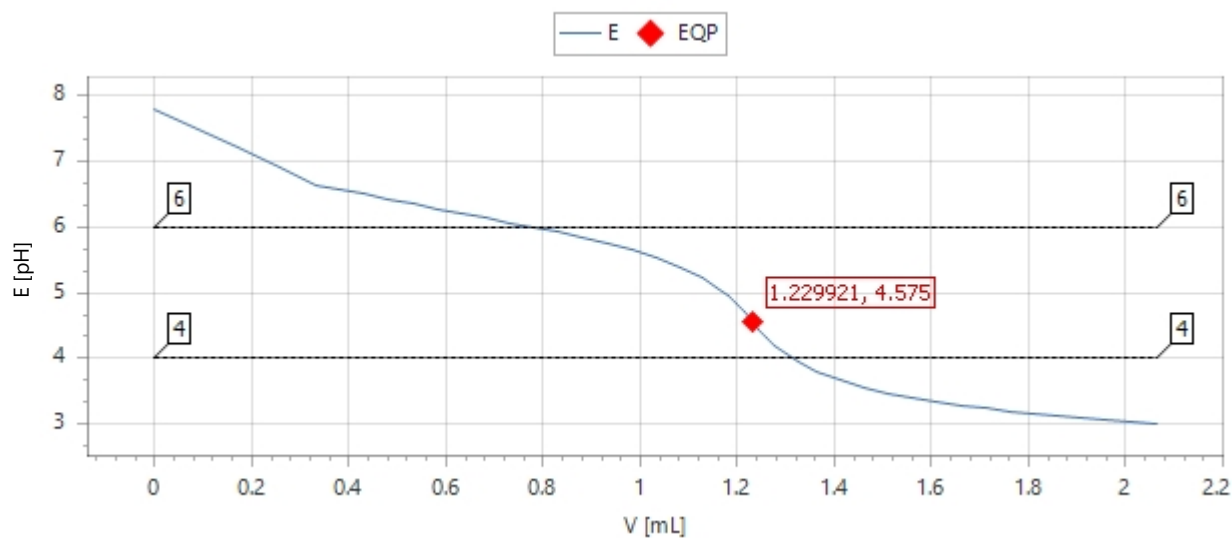
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39	7.641	25.3	0.015
40	7.656	25.3	0.015
41	7.669	25.3	0.013
42	7.681	25.3	0.012
43	7.692	25.3	0.011
44	7.702	25.3	0.010
45	7.711	25.3	0.009
46	7.720	25.3	0.009
47	7.729	25.3	0.009
48	7.737	25.3	0.008
49	7.743	25.3	0.006
50	7.750	25.3	0.007
51	7.756	25.3	0.006
52	7.762	25.3	0.006
53	7.767	25.3	0.005
54	7.772	25.3	0.005
55	7.776	25.3	0.004
56	7.781	25.3	0.005
57	7.784	25.3	0.003
58	7.788	25.3	0.004
59	7.792	25.3	0.004
60	7.795	25.3	0.003

Titration (EQP) [1] Scope 1/1, Sample 15/17

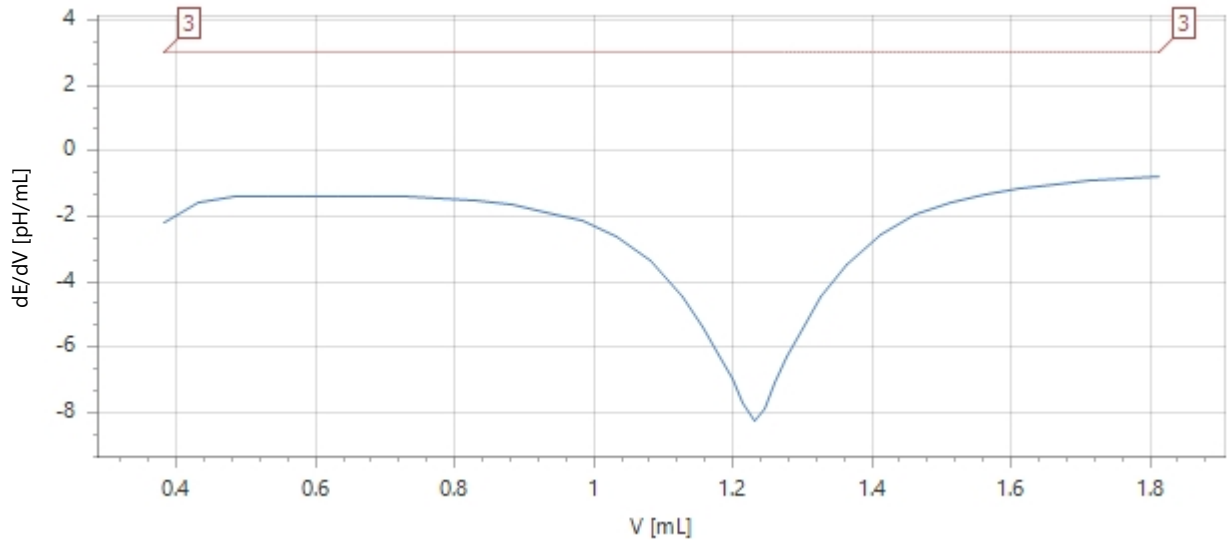
E - V Curve Scope 1/1, Sample 15/17



dE/dV - V Curve Scope 1/1, Sample 15/17

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Measured Values Scope 1/1, Sample 15/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.803		25.3			
29	0.17150	7.227	0.17150	25.3	-0.576		
59	0.25725	6.918	0.08575	25.3	-0.309		
90	0.30000	6.753	0.04275	25.2	-0.165		
181	0.33250	6.649	0.03250	25.2	-0.104		
211	0.38250	6.578	0.05000	25.2	-0.071	-2.19	
241	0.43250	6.499	0.05000	25.2	-0.079	-1.57	
271	0.48250	6.423	0.05000	25.2	-0.076	-1.36	
302	0.53250	6.350	0.05000	25.2	-0.073	-1.39	
332	0.58250	6.277	0.05000	25.2	-0.073	-1.42	
362	0.63250	6.208	0.05000	25.2	-0.069	-1.39	
392	0.68250	6.140	0.05000	25.2	-0.068	-1.38	
422	0.73250	6.070	0.05000	25.2	-0.070	-1.41	
453	0.78250	5.997	0.05000	25.2	-0.073	-1.46	
483	0.83250	5.922	0.05000	25.2	-0.075	-1.54	
513	0.88250	5.842	0.05000	25.2	-0.080	-1.66	
544	0.93250	5.753	0.05000	25.2	-0.089	-1.85	
574	0.98250	5.652	0.05000	25.2	-0.101	-2.15	
604	1.03250	5.537	0.05000	25.2	-0.115	-2.62	
634	1.08250	5.395	0.05000	25.2	-0.142	-3.38	
664	1.12800	5.229	0.04550	25.2	-0.166	-4.48	
695	1.15700	5.093	0.02900	25.2	-0.136	-5.40	
725	1.17975	4.962	0.02275	25.2	-0.131	-6.23	
755	1.19850	4.832	0.01875	25.2	-0.130	-6.96	
785	1.21500	4.702	0.01650	25.2	-0.130	-7.70	
	1.229921 ⁽⁹⁾	4.575 ⁽⁹⁾					EQP
816	1.23025	4.572	0.01525	25.2	-0.130	-8.23	
846	1.24475	4.446	0.01450	25.2	-0.126	-7.90	
876	1.26025	4.321	0.01550	25.2	-0.125	-7.08	
906	1.27850	4.191	0.01825	25.2	-0.130	-6.28	
937	1.29975	4.065	0.02125	25.2	-0.126	-5.45	

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967	1.32750	3.935	0.02775	25.2	-0.130	-4.46
997	1.36400	3.803	0.03650	25.2	-0.132	-3.45
1027	1.41250	3.668	0.04850	25.2	-0.135	-2.53
1058	1.46250	3.561	0.05000	25.2	-0.107	-1.94
1088	1.51250	3.475	0.05000	25.2	-0.086	-1.57
1118	1.56250	3.402	0.05000	25.2	-0.073	-1.32
1148	1.61250	3.340	0.05000	25.2	-0.062	-1.14
1178	1.66250	3.285	0.05000	25.2	-0.055	-1.02
1209	1.71250	3.237	0.05000	25.2	-0.048	-0.91
1239	1.76250	3.193	0.05000	25.2	-0.044	-0.83
1269	1.81250	3.153	0.05000	25.2	-0.040	-0.76
1300	1.86250	3.117	0.05000	25.2	-0.036	
1330	1.91250	3.083	0.05000	25.2	-0.034	
1360	1.96250	3.052	0.05000	25.2	-0.031	
1390	2.01250	3.023	0.05000	25.2	-0.029	
1420	2.06250	2.996	0.05000	25.2	-0.027	

Used Instruments

Name	SERCtitrate	Serial number	B433917554
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Responsible User

Time	Login name	Full name
2/22/2024 12:10:04 PM	SystemInternal	System internal account

(9) Standard evaluation used
(12) Unfinished sample